

Concerns and Challenges of AI Tools in the UI/UX Design Process: A Cross-Sectional Survey

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ABSTRACT

The study aimed to gain a comprehensive understanding of AI integration in UI/UX design process from the perspective of professionals directly engaged with these tools. Two surveys were conducted in August 2023, to explore attitudes, ethical concerns, and future views on AI-assisted design technologies through both closed and open-ended questions. Using both thematic and descriptive analyses, the results showed that designers were cautiously optimistic about AI tools. This shows how important it is to think about ethics, understand how AI works, and find a good balance between integrating AI into the design workflow. These insights are crucial for AI tool developers, design educators, and the design community to develop more effective, ethical, and user-centric AI tools.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User studies**;

KEYWORDS

artificial intelligence, design process, UX, professionals, survey, tools, recommendations

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1 INTRODUCTION

The field of user interface and user experience (UI/UX) design is undergoing a transformative phase with the recent advancements in artificial intelligence (AI) [3]. AI tools and technologies are increasingly being used to automate repetitive tasks, generate design ideas, and provide data-driven insights. This has not only led to an increase in the efficiency and effectiveness of the design process [1, 16] but also introduced new paradigms in the creative process of design [8, 10]. As AI becomes more sophisticated, its role in design will expand, leading to new opportunities and challenges [20, 21].

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While the technological aspects of AI in design are rapidly expanding, there is a need to understand the human side of this integration, particularly the perspectives of UI/UX designers who are the end-users of these AI tools. Only a few studies have explored this topic, and they are limited by cultural contexts [12] or sample sizes [9]. Additional research can help increase the understanding of AI-assisted UX design and fine-tune the published recommendations.

We conducted a study to gather comprehensive insights from UI/UX designers about their experiences, concerns, and challenges with AI tools in their professional practice. This included gaining an insight into their perceptions of the benefits and drawbacks of AI in design, its impact on their workflow and creative process, and the need for additional skills or training. In addition, we aimed to identify ethical concerns and associated barriers to the effective integration of AI in UI/UX the design profession.

Our findings contribute to a more nuanced understanding of AI's role in UI/UX design from the perspective of the designers themselves. AI tool developers, design educators, and the design community at large will find these results valuable as they inform the development of more effective, ethical, and designer-centered AI tools. Furthermore, these findings help prepare current and future designers for a rapidly evolving design landscape infused with AI.

2 METHODOLOGY

The aim of this study was to gain a comprehensive understanding of the various aspects of AI integration in UI/UX design from those who are directly engaged with these tools in their professional environment. We conducted two surveys with UI/UX design professionals over a period of two weeks in August 2023. Ethical approval was secured from the institutional review board of our home institution prior to the launch of the surveys.

Participants were recruited by posting the surveys in online professional groups for UI/UX designers. To be eligible, participants needed to be currently working in a UI/UX role at a company, have at least one year of experience in UI/UX design, and have used or are familiar with AI-assisted UI/UX tools. This selection process was designed to ensure that respondents could provide valuable and informed perspectives about the use of AI in design.

The survey did not collect any identifying information. Respondents were free to answer as many questions as they wanted and quit the survey at any time. To encourage voluntary participation based on interest in the subject matter rather than incentivization, no remuneration was offered for participating in the surveys.

2.1 Participants

73 ($n_1 = 45$, $n_2 = 28$) UI/UX design professionals completed both surveys. 25 ($f_1 = 15$, $f_2 = 10$) participants were female, and the average age of the participants was 32.4 years. Participants had designed UX products for different industries such as health care, education, finance, etc. The average work experience of the participants was 5 years ($Y_1 = 5.8$ years, $Y_2 = 4.3$ years). Participants reported using three types of AI design tools: text-based generative AI tools (28%), such as ChatGPT and AutoGPT; intelligent prototyping tools (34%), such as Sketch and ProtoPie; and image manipulation and generation tools (45%), such as MidJourney and Huemint.

2.2 Survey Design

The first part of both surveys consisted of a brief description of the study, inclusion criteria, and informed consent. Participants were then asked to provide the necessary demographics (e.g., gender, age, years of experience, job title, current industry). The second part of both surveys concentrated on specific questions related to the study goals. In the first survey, participants were asked 8 questions, comprising one close-ended and 7 open-ended questions. The close-ended question was a Likert-scale question, consisting of 8 items, where each was anchored on a scale ranging from 1 (strongly disagree) to 5 (strongly agree). The open-ended questions required qualitative responses. Overall, the survey questions fell into three primary categories: attitudes, ethical concerns, and future speculations about AI-assisted design technologies. The second survey included three open-ended questions aimed at identifying the topics, skills and training needs of UI/UX design professionals regarding AI-based design tools.

2.3 Data Analysis

Microsoft Excel was employed to perform a descriptive analysis of the closed questions, involving the calculation of means, standard deviations (SDs), and frequency distributions. This facilitated a comprehensive statistical overview of the quantitative data collected from the survey.

We conducted a thematic analysis, following the approach outlined by Braun and Clarke [4], to delve deeply into the open-ended responses provided by UI/UX design professionals. We utilized both inductive and deductive coding techniques, such as open coding and memoing, to identify initial patterns and themes related to AI-assisted design tools.

Initially, the research team engaged in an extensive review of each response, ensuring a deep understanding of the designers' perspectives on using AI in their workflows. This phase allowed us to generate initial codes that captured the core sentiments and viewpoints expressed by the participants, focusing specifically on how AI tools are being integrated into their design practices.

The analysis then proceeded to organize these initial codes into more defined subthemes, carefully reflecting the detailed experiences of UI/UX designers with AI-assisted tools. This iterative refinement process was directly aligned with the study's primary objective: to explore the complex landscape of designers' perceptions, the challenges they encounter, and the ethical considerations they weigh regarding the use of AI in design.

By systematically refining these subthemes and grouping them into broader categories, we synthesized the varied perspectives into three, coherent thematic domains. This step not only highlighted the complex nature of designers' experiences and concerns about AI but also clarified the overarching patterns and insights regarding the integration of AI technologies in the design process.

The outcome of this analytical journey is a thematic framework that captures a detailed understanding of UI/UX designers' attitudes towards AI, the challenges they face, and their ethical deliberations about these tools in design.

3 RESULTS

3.1 Subjective attitudes towards AI-assisted UI/UX design

The close-ended Likert scale questions regarding designers' attitudes towards AI-assisted design was added after 8 participants had already completed the survey, therefore, only 37 participants answered this question. For two items, only 36 responses were recorded. The responses are summarized in Figure 1. Based on these data, it can be inferred that, overall, participants had positive attitudes towards AI-assisted design tools ($mean = 3.71$; $SD = 0.90$). Most participants believed that these tools enhanced the designers' performance (items 1, 2) and the quality of UI/UX (item 3). They also believed that AI-assisted tools would improve their communication with stakeholders (item 4). Most believed more strongly that AI-assisted tools should be designed to support collaboration (item 8) and inspiration (item 7) activities involved in design work as opposed to automation of design tasks (item 6). However, the majority of participants exhibited neutral to strong concerns about future job displacement due to AI ($SD = 1.18$) as indicated in item 5, reflecting apprehensions about AI potentially taking over their job roles.

3.2 Experiences, concerns, and challenges with AI-assisted design tools

The qualitative analysis led to the identification of three main themes that help us understand the challenges participants perceived in terms of working with AI-assisted design tools. We describe the related subthemes below.

3.2.1 AI's Impact on Creativity and Design skills

(a) *AI may stifle innovation.* Concerns were raised about the quality of AI-generated designs, particularly regarding their perceived lack of innovation. "One concern is that it will ultimately stop being creative because it lacks human-like judgment," noted P34. This sentiment reflected a broader apprehension that, while AI can efficiently generate design elements based on existing data and trends, it may falter in creating truly innovative or out-of-the-box designs that push creative boundaries. Participants highlighted a potential consequence of this limitation: "a homogenization of design aesthetics," (P11). Such a trend could result in AI-generated designs that lack distinctiveness and fresh perspectives, leading to a landscape where designs feel overly similar and lack the unique spark that human creativity can ignite.

(b) *Continuous usage may deteriorate foundational design skills.* Participants highlighted several challenges and potential risks for

	1	2	3	4	5	Mean	Mode	Median	SD
1. improve my overall performance.	2	0	4	25	6	3.89	4	4	0.88
2. increase my effectiveness in UI/UX tasks.	1	1	7	21	7	3.86	4	4	0.86
3. improve the quality of UI/UX design processes.	1	0	7	23	6	3.89	4	4	0.77
4. enhance communication with the stakeholders.	1	2	7	23	4	3.73	4	4	0.84
5. would replace my job in the future.	4	7	10	12	4	3.14	4	3	1.18
6. should be used for automating UI/UX design processes.	1	3	13	14	5	3.53	4	4	0.94
7. should be used to provide inspiration for UI/UX design work.	1	0	9	22	5	3.81	4	4	0.78
8. should be designed to collaborate with designers.	0	2	7	23	4	3.81	4	4	0.71

Figure 1: Likert scale of participants' attitude towards AI-assisted design tools. Each item begins with the following clause: "I believe that AI-assisted design tools ..."

designers. Specifically, they expressed concerns that over-reliance on AI tools could lead to a decrease in proficiency with fundamental design principles and practices as automated processes take over tasks that designers traditionally performed manually. *"It poses a risk to unlearn the self-creation capabilities over time,"* (P27). They expressed concern that AI-shift could potentially lead to a skill gap in the industry, where upcoming designers might lack expertise in certain core areas of UI/UX design.

3.2.2 The Necessity of Human-Centric AI Integration

(a) *AI design tools should prioritize human oversight and control.* Participants expressed a strong desire for greater control over AI-generated content, emphasizing the critical role of human oversight in AI decision-making processes. They argued that while AI can significantly enhance design capabilities, final decisions should rest with human designers, who are best positioned to contextualize and evaluate AI suggestions to ensure alignment with user needs and ethical considerations. *"They must adhere to the principle that humans retain ultimate control over the decisions made by the AI,"* (P8) underscored this point. Additionally, there was a consensus on the necessity for AI tools to explain their decision-making processes, making the AI's logic more accessible and comprehensible to human designers. P26 highlighted the importance of developing transparent and user-friendly interfaces that allow designers to fine-tune AI outputs and adjust parameters to effectively meet specific design goals.

(b) *Designers must acquire new skills to use AI tools.* The need to continuously update and acquire new skills to keep pace with rapidly evolving AI technologies was identified as a significant challenge. Many participants highlighted the need for ongoing education and training to stay current with the latest advancements in AI tools. This requirement, however, was noted as being particularly demanding in terms of both time and resources. The designers also revealed the need to do more than just operate AI tools; they also indicated that they must also understand the underlying principles of these technologies to effectively integrate them into their work. *"It has created the need to learn, understand, and develop measures to address potential risks associated with AI integration into the design process. This is in addition to our daily UX work,"* (P22). The respondents described this learning curve as daunting and time-consuming, often detracting from their capacity to focus on other essential design tasks.

(c) *AI tools may not be compatible with existing design workflows.* Many respondents raised concerns about integrating AI tools into their established design processes. They feared that existing workflows may not seamlessly accommodate AI technologies, requiring adjustments or even complete overhauls of traditional methods. This concern was particularly pronounced regarding the compatibility of AI tools with current software and systems, as well as the potential disruption to team dynamics and established routines. *"A big question is about the impact of AI on design system teams. Will they become smaller and leaner as AI gets more prevalent?"* (P17). Respondents were apprehensive about the time and resources needed to adapt to new AI-driven methodologies, worrying that this could lead to declines in productivity and efficiency.

3.2.3 Technical and Ethical Challenges of AI Deployment

(a) *AI cannot replicate human capabilities.* Participants expressed concerns about AI's inability to replicate human traits such as creativity, subjectivity, and emotional depth. They pointed out that AI's reliance on data does not suffice in scenarios demanding human intuition and emotional intelligence. One participant noted, *"Design is a creative and subjective work, involving personal aesthetics and emotions. AI systems often struggle to understand and simulate this kind of creative human thinking and emotional expression, so AI's role may be limited when it comes to design decisions and idea generation,"* (P12). Designers emphasized the importance of human-centric elements in creating designs that resonate deeply with users. Additionally, concerns were raised about AI's inability to comprehend and interpret the subtle complexities of human experiences and cultural contexts—factors considered crucial for developing empathetic and engaging user experiences. P27 explained, *"In my experience, cultural context is crucial in design and content creation. What works in one culture can offend in another, and AI often misses these nuances, leading to significant missteps."*

(b) *Sufficient and appropriate training data is challenging to acquire.* The reliance on extensive datasets for AI training was a significant concern, particularly regarding data quality and potential biases. Designers noted that the accuracy and impartiality of these datasets are crucial, as any inherent biases would result in designs that do not adequately represent or appeal to diverse user groups. *"AI models learn based on training data, and if training data is biased or discriminatory, then the model may reflect these biases and introduce inequities or injustices into the design,"* remarked P16.

Participants indicated that meticulous data collection and analysis practices are essential for mitigating the risk of perpetuating stereotypes or cultural biases through AI-generated outputs. However, they also acknowledged significant challenges in amassing large, diverse datasets for AI. The rapid evolution of digital interfaces necessitates frequent updates to ensure datasets remain relevant, a task P34 described as continually challenging due to the fast-paced nature of technology. Additionally, P42 pointed out the difficulty of merging data from various platforms and formats, underscoring the complexity of creating cohesive datasets for effective AI training. These challenges underscore the intricate process of developing AI models that are relevant to users.

(c) *AI algorithms have the potential to harm the user.* The critical importance of algorithmic integrity for achieving fair and unbiased AI applications was a prominent concern among participants. They emphasized that biases embedded within algorithms could not only mirror but potentially worsen societal inequities through design. *"Without fair algorithms and applications, it is not fair to use AI to aid design, and that is the challenge we face,"* noted P33, highlighting the ethical stakes involved. Participants underscored that striving for algorithmic fairness transcends technical hurdles, representing a moral obligation to ensure AI acts as a force for good in society. They advocated for the development and implementation of methodologies to detect and neutralize such biases in existing algorithms. Yet, they also recognized that the task of purging algorithms of bias presents significant challenges. *"Achieving truly unbiased algorithms is challenging; it is about confronting and correcting the biases that have historically influenced our data and, by extension, our AI systems,"* remarked P15. This discussion points to the dual necessity of vigilance and innovation in refining AI algorithms, ensuring they serve to bridge rather than broaden societal gaps.

(d) *AI poses a threat to privacy and confidentiality.* Participants voiced worries about the privacy and confidentiality of users' information collected for analysis and requirements analysis purposes. *"There is a potential risk of information leakage whenever data is fed into AI systems,"* (P6) said. They expressed concerns over the potential misuse of sensitive user data, highlighting the ethical responsibilities of designers and developers in ensuring data security and user privacy. Moreover, P27 highlighted the delicate act of gathering data for training AI algorithms without compromising user privacy, emphasizing the importance of user consent and anonymity, which was currently being ignored in the development of AI algorithms. There was a consensus on the importance of implementing robust data protection measures and transparent data handling practices. The participants called for strict adherence to privacy laws and regulations, along with the incorporation of ethical guidelines in AI development and deployment.

3.3 Learning needs around AI-assisted design technologies

Topics that participants were interested in investigating about AI-assisted design tools have been summarized in Table 1, along with specific questions to consider. We describe these learning needs in detail below.

3.3.1 Understanding the Scope and Limitations of AI in Design. Designers need a clear understanding of what AI can and cannot do

within the design process. This includes recognizing areas where AI cannot replicate human creativity, subjectivity, and emotions, and identifying when human intervention is crucial for creativity and innovation. Additionally, they must learn to manipulate AI to prevent homogenization and ensure that designs remain innovative and diverse.

3.3.2 Gaining Technical Proficiency in AI Tools and Algorithms. Designers need a clear understanding of what AI can and cannot do within the design process. This includes recognizing areas where AI cannot replicate human subjectivity, and emotions, and identifying when human intervention is crucial for creativity and innovation. Additionally, they must learn to manipulate AI to prevent homogenization and ensure that designs remain innovative and diverse.

3.3.3 Learning Ethical Use of AI in Design. With concerns over privacy, data misuse and potential biases in AI, designers need to be well-versed in data protection measures and ethical guidelines. They can benefit from training on data privacy laws, ethical design principles, and practical strategies for implementing robust data protection in AI-assisted design projects. It is also crucial for designers to understand and learn how to navigate data bias concerns to ensure designs do not inadvertently harm or exclude user groups.

3.3.4 AI in Team and Project Management. Designers envision that AI will reshape the existing workflows. It is crucial that designers learn strategies for incorporating AI into established design workflows, all while ensuring productivity and team dynamics remain undisturbed. In addition, learning the use of AI beyond traditional design tasks to include team dynamics, task management, and adapting to constant changes in project requirements is an emerging need of designers. Moreover, designers envision that AI will carve out new roles within the design industry, such as AI trainers, prompt designers, and augmented reality designers. They indicated requiring additional training to learn these goals.

3.3.5 Navigating the Legal Landscape of AI in Design. With the legal ramifications of using AI tools becoming a growing concern, designers need to be aware of the legal aspects, including liability issues, patent eligibility, and compliance with data protection laws. They need to learn how to protect intellectual property and navigate liability when AI tools are involved in the design process.

4 DISCUSSION

This study aimed to investigate the current perceptions and challenges associated with AI-assisted tools in the context of UI/UX design. The results show a cautious but insightful perspective from designers regarding the integration of AI in UI/UX design. These insights call for further research into achieving a balanced integration of AI in UI/UX design by being mindful of its limitations and impacts on UX design process and designers. The development of AI tools that meet ethical standards and empower designers is also highlighted. Furthermore, this work highlights certain areas where designers would benefit from additional training. We discuss three major implications of the study findings.

Table 1: AI-related learning needs of UI/UX designers

Topic	Sample Questions
Scope and Limitations of AI in Design	• What limitations should UX designers be aware of when using AI in their projects? (P3) • When to use and when to never use AI in your design process? (P18)
Technical Knowledge of AI Algorithms	• How AI makes decisions, so we know when to accept or reject its suggestions? (P12) • How can I train AI in specific scenarios to get valuable insights on design process? (P1)
Ethical Use of AI in Design	• How to ensure protection of user data, especially when submitting user interviews and sensitive user data through ChatGPT? (P3) • What are your tactics for watching out for hallucination, bias, and other blind spots when designing with AI? (P14)
AI in Team and Project Management	• How can we use AI for better managing a UX team, mostly task management and change management, along with the product team's constant needs for changes? (P21)
Legal Landscape of AI in Design	• What are the legal ramifications of designing with AI? (P2) • If a system built on AI results in harm, who is responsible? (P11) • Will solutions assisted by AI be eligible for patents? (P23)

4.1 Support empathy and creativity of designers

The existing literature highlights the current limitations of AI in achieving true artistic creativity, particularly its struggle to fully understand the complexities of human abilities across various domains [2, 11]. This observation is in line with concerns raised by study participants about AI's inability to replicate the essential human creativity and intuition required for UX design tasks. Research has explored AI's varied roles in creative processes, from acting as self-sufficient generators to serving as tools that augment human creativity [6, 19]. Our findings lean towards the latter, suggesting that future AI tools in design should not only enhance but also support the creativity and empathy of designers. Rather than generating an output [5], these tools should act as active partners in the creative process, learning from and adapting to the feedback and preferences of designers. AI's contributions should be nuanced and closely aligned with the designer's style and the specific needs of each project, potentially improving AI's capacity to emulate human capabilities in creative design endeavors.

By being more responsive to designers' thought processes and allowing for the integration of their own data, this approach could also address some ethical concerns associated with AI-assisted design tools. By preserving the unique and irreplaceable elements of human creativity and empathy in design, it could potentially safeguard against the overstandardization of AI creativity that might stifle innovation and creative potential. Ultimately, AI-assisted creativity should complement human limitations, focusing on enhancing human creativity within the design process.

4.2 Enhance ethical sensibilities of AI-Assisted design tools

Inherent biases in AI algorithms have become a critical concern among designers, who worry that AI might produce biased or misrepresented design outputs due to the training on biased data. Consequently, enhancing the ethical sensibilities of AI-assisted design tools is a crucial priority.

Addressing this requires a comprehensive approach, including diversifying training data to cover a broad range of demographics and backgrounds, thereby reducing bias risks [17]. This should be accompanied with regular audits to identify and correct biases in AI systems [14]. Cross-cultural collaboration can offer diverse ethical perspectives, enhancing the design process. Additionally, a multi-stakeholder approach to AI governance, involving collaboration among designers, ethicists, legal experts, and user representatives, is crucial [15].

Supporting ethical guidelines such as fairness, inclusivity, and transparency should be a priority for AI tool development. At the interface level, tools should be able to support human oversight throughout the design process to ensure alignment with human values and ethics [13]. Furthermore, innovative tools and interfaces that support ethical prompt engineering for design tasks could be developed.

4.3 Integrate AI into existing workflows as an intelligence layer to design systems

To address designers' concerns about AI tool compatibility with existing workflows, developing smarter, modular design systems with an integrated 'intelligence layer' is key [18]. This layer would act as an assistant in the UX design process, offering insights into user interactions, such as navigation patterns and time spent on various page elements. It would also suggest effective design elements and functions, and identify unintentional biases, aiding in the creation

of inclusive designs for a diverse user base. The intelligence layer's primary role is to manage analytical and data processing tasks, freeing designers to focus on creativity and critical design aspects.

Integrating this layer could resolve compatibility issues with AI tools in design processes, transforming them from potentially disruptive separate entities into cohesive components of the design system. It could preserve team dynamics and workflows [7], while leveraging AI's benefits to enhance productivity and efficiency. User behavior insights would enable designers to make informed decisions, resulting in better-designed products and faster design iterations.

4.4 Addressing learning needs of UX designers

An in-depth analysis of identified learning needs could reveal their impact on design quality and highlight areas where AI knowledge gaps have led to issues in past projects. Developing a targeted curriculum that includes workshops, ethical design seminars, and courses on AI algorithms would address these needs directly. Continuous professional development programs could keep designers updated on AI advancements and ethical standards. Finally, exploring future research directions, such as the effectiveness of learning approaches and the impact of AI literacy on design, would provide valuable insights into further integrating AI in the design process.

5 CONCLUSION

UX designers regard AI-assisted design tools as significant and evolving components of the design process, yet they express concerns about the limitations, ethical implications, and impact of these tools on both designers and the design process. Future development of AI-assisted design tools should prioritize supporting designers' empathy and creativity, enhancing the tools' ethicality, and integrating AI-enabled intelligence layers within the existing design systems to minimize disruptions in workflows.

REFERENCES

- [1] Sumeen Agrawal, Manith Shetty, Sripurna Mutalik, and Anuradha Kanukotla. 2022. Method to Improve UI Rendering using Predictive Sequence Modelling. In *2022 26th International Conference on Pattern Recognition (ICPR)*. IEEE, 5031–5037.
- [2] Gérard Assayag. 2016. Improvising in creative symbolic interaction. In *Mathematical conversations: Mathematics and Computation in Music Performance and Composition*. World Scientific, 61–74.
- [3] Renato Antonio Bertão and Jaewoo Joo. 2021. Artificial intelligence in UX/UI design: a survey on current adoption and [future] practices. *Safe Harbors for Design Research* (2021), 1–10.
- [4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [5] Francisco Maria Calisto, João Fernandes, Margarida Morais, Carlos Santiago, João Maria Abrantes, Nuno Nunes, and Jacinto C Nascimento. 2023. Assertiveness-based agent communication for a personalized medicine on medical imaging diagnosis. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–20.
- [6] Dominik Dellermann, Adrian Calma, Nikolaus Lipusch, Thorsten Weber, Sascha Weigel, and Philipp Ebel. 2021. The future of human-AI collaboration: a taxonomy of design knowledge for hybrid intelligence systems. *arXiv preprint arXiv:2105.03354* (2021).
- [7] KJ Kevin Feng, Tony W Li, and Amy X Zhang. 2023. Understanding Collaborative Practices and Tools of Professional UX Practitioners in Software Organizations. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–20.
- [8] Xinyu Guo, Yi Xiao, Jiaqi Wang, and Tie Ji. 2023. Rethinking designer agency: A case study of co-creation between designers and AI. (2023).
- [9] Tiffany Kneare, Mohammed Khwaja, Yuling Gao, Frank Bentley, and Clara E Kliman-Silver. 2023. Exploring the future of design tooling: The role of artificial intelligence in tools for user experience professionals. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [10] A Baki Kocaballi. 2023. Conversational ai-powered design: Chatgpt as designer, user, and product. *arXiv preprint arXiv:2302.07406* (2023).
- [11] Arthur I Miller. 2019. *The artist in the machine: The world of AI-powered creativity*. Mit Press.
- [12] Prasadini Padmasiri, Pramukthika Kalutharage, Nethma Jayawardhane, and Jagath Wickramaratne. 2023. AI-Driven User Experience Design: Exploring Innovations and Challenges in Delivering Tailored User Experiences. In *2023 8th International Conference on Information Technology Research (ICITR)*. IEEE, 1–6.
- [13] Trishan Panch, Heather Mattie, and Rifat Atun. 2019. Artificial intelligence and algorithmic bias: implications for health systems. *Journal of global health* 9, 2 (2019).
- [14] Inioluwa Deborah Raji and Joy Buolamwini. 2019. Actionable auditing: Investigating the impact of publicly naming biased performance results of commercial ai products. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. 429–435.
- [15] Reva Schwartz, Apostol Vassilev, Kristen Greene, Lori Perine, Andrew Burt, Patrick Hall, et al. 2022. Towards a standard for identifying and managing bias in artificial intelligence. *NIST special publication* 1270, 10.6028 (2022).
- [16] M SureshKumar, G Annamalai, S Kishore Kumar, and N Susila. 2022. AI Based UI Development Using Deep Neural Networks (AI-UI). In *2022 1st International Conference on Computational Science and Technology (ICCST)*. IEEE, 387–391.
- [17] Daniel Varona and Juan Luis Suárez. 2022. Discrimination, bias, fairness, and trustworthy AI. *Applied Sciences* 12, 12 (2022), 5826.
- [18] Dieter P Wallach, Lukas A Flohr, and Annika Kaltenhauser. 2020. Beyond the buzzwords: On the perspective of ai in ux and vice versa. In *International Conference on human-computer interaction*. Springer, 146–166.
- [19] Dakuo Wang, Elizabeth Churchill, Pattie Maes, Xiangmin Fan, Ben Shneiderman, Yuanchun Shi, and Qianying Wang. 2020. From human-human collaboration to Human-AI collaboration: Designing AI systems that can work together with people. In *Extended abstracts of the 2020 CHI conference on human factors in computing systems*. 1–6.
- [20] Bai Yang, Ying Liu, and Wei Chen. 2023. A twin data-driven approach for user-experience based design innovation. *International Journal of Information Management* 68 (2023), 102595.
- [21] Zhibin Zhou, Lingyun Sun, Yuyang Zhang, Xuanhui Liu, and Qing Gong. 2020. ML lifecycle canvas: designing machine learning-empowered UX with material lifecycle thinking. *Human-Computer Interaction* 35, 5-6 (2020), 362–386.